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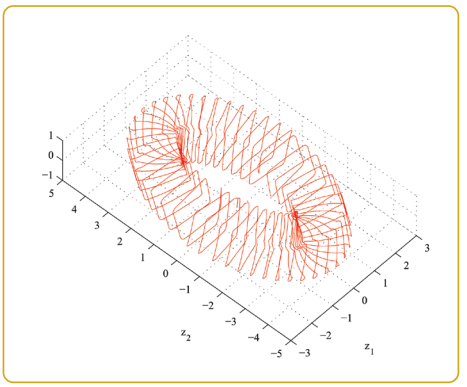
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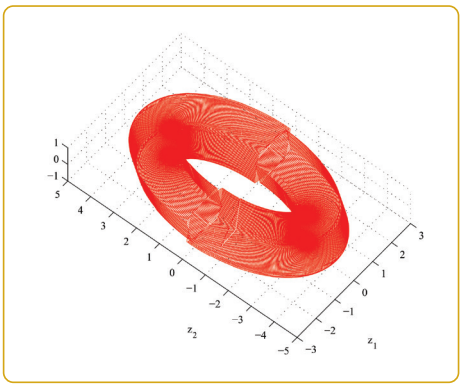
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Averaging of Singularly Perturbed Control Systems and LP Approach to Optimal Control





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In memory of Anatolii Arkadievich Pervozvanskii

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Preface

The whole is greater than the sum of its parts.
—Aristotle

This book represents a long line of research on the averaging of singularly perturbed (SP) control systems and the linear programming (LP) approach to optimal control (OC). Our primary aim is to provide a systematic exposition of averaging techniques for SP control systems, with particular emphasis on their interplay with infinite-dimensional LP (IDLP) formulations of OC problems.

Early research on SP systems often relied on setting the singular perturbation parameter to zero and replacing the fast state variables with their "quasi steady state" values. While effective in some contexts, this method fails to capture the influence of rapidly oscillating controls, which induce oscillations in the fast state variables and significantly affect the slow dynamics.

An adequate approach to capturing such rapid oscillations is averaging. In this framework, the slow state trajectories are approximated by the averaged differential inclusion (ADI), whose right-hand side is defined as the limit of the time averages of the slow dynamics over the control-state trajectories of the associated system (the fast subsystem with frozen slow variables). This limit exists and is convex whenever the associated system satisfies a mixing condition, itself guaranteed under suitable stability or controllability assumptions.

A stronger version of the mixing condition ensures the existence and convexity of the limit occupational measures set (LOMS), generated by the control-state trajectories of the associated system. This makes it possible to define an alternative averaged model—the averaged control system (ACS)—in which the LOMS acts as the control set. Under additional conditions, the LOMS coincides with the set of stationary measures of the associated system, which is explicitly described by linear equations.

The solution sets of the ADI and the ACS coincide. In some situations, the ADI is the more natural tool; in others, the ACS provides a more effective framework. Notably, the fact that the LOMS coincides with the set of stationary measures (and is therefore defined by linear equations) allows one to apply LP-based techniques to design near-optimal controls for the ACS, which can then be used to construct asymptotically near-optimal controls for the SP system.

The book consists of two parts. In Part I, we introduce the ADI and the ACS, provide conditions under which the limit of the set of time averages and the LOMS exist and are convex, and discuss conditions under which the LOMS coincides with the set of stationary measures. We also demonstrate how averaging applies to multi-time-scale systems and systems with slow observables. In Part II, we develop the LP approach to OC problems with time-averaging and time-discounting criteria. We establish strong duality results for IDLP problems, related to the OC problems under consideration; derive sufficient and necessary optimality conditions based

on these duality results; and study semi-infinite-dimensional (SID) approximations of the IDLP problems, whose solutions enable the construction of near-optimal controls. Numerical methods for solving these approximations are also discussed. Taken together, the results of the two parts provide both conceptual foundations and constructive tools for designing asymptotically optimal and near-optimal controls of SP systems.

Many of the results presented in the book first appeared in research articles addressing specific problems or techniques. Taken individually, these works illuminated particular aspects of the theory. We hope that, brought together here, they offer a more comprehensive and coherent framework, clarifying the interconnections between methods and revealing a deeper structure.

This book is intended for graduate students, researchers, and practitioners working in control, optimization, and applied mathematics. While it assumes some familiarity with functional analysis, measure theory, and optimal control, the presentation is designed to remain largely self-contained.

I would like to express my deep gratitude to my coauthors, whose ideas and contributions are interwoven throughout this book (their names appear on the joint papers listed in the references). In particular, I would like to acknowledge Sergey Rossomakhine, who conducted a number of elaborate numerical experiments illustrating the theory. The results of these experiments were published in joint papers, and some of them are presented in this book. I am also very grateful to colleagues with whom I have not coauthored papers, but whose insights, discussions, and valuable comments have influenced my work—among them Jean-Pierre Aubin, Martino Bardi, Fritz Colonius, the late Asen Dontchev, Paul Dupuis, Hélène Frankowska, Harold Kushner, Terry Rockafellar, and Vladimir Veliov.

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Chapter 1
Introduction

1.1 ▪ Background and Overview

The ever-increasing range of applications of nonlinear controlled dynamical systems, along with the growing complexity of the models used (driven by the desire to make them more representative of real-life systems), has led to steadily rising interest in this field among both practitioners and mathematicians. A major difficulty in working with complex controlled dynamical systems is the disparity of time scales, often manifested through the presence of a singular perturbation parameter or parameters. These parameters lead to a separation of state variables into groups that evolve at significantly different rates, allowing some variables to be classified as "fast" or "slow" relative to others.

The most common approach to singularly perturbed (SP) control systems is the order reduction technique, based on setting the singular perturbation parameter to zero and thereby replacing the fast state variables with their "quasi steady state" values. This approach has received a great deal of attention in the literature, with a bibliography that includes hundreds of papers as well as a number of research monographs and surveys (see, for example, the surveys by Dmitriev and Kurina [56], Dmitriev et al. [57], Kokotović [130], Naidu [142], and Zhang et al. [176], and the research monographs by Bensoussan [38], Dontchev [61], Dontchev and Zolezzi [65], Kabanov and Pergamenshchikov [126], Kokotović, Khalil, and O'Reilly [131], Kushner [133], O'Malley [144], and Yin and Zhang [173]).

While highly effective for many important classes of optimal control (OC) problems, the order reduction technique may not be applicable in the general case. Specifically, the assertion that the system obtained by setting the singular perturbation parameter to zero (commonly referred to as the reduced system) can be used to find a near-optimal control for the SP system relies on the hypothesis that the optimal control is, in some sense, slow, and that the fast state variables rapidly converge to their quasi steady states and remain near them, while the slow variables evolve according to the reduced dynamics. However, this hypothesis may not hold in the control setting if the dynamics are nonlinear and/or the objective function is nonconvex. In such cases, the use of rapidly oscillating controls can lead to substantial improvements in the performance indexes, improvements that do not vanish as the singular perturbation parameter tends to zero.

This book is devoted to the study of averaging and linear programming (LP)-based techniques applicable to the analysis of OC problems for SP systems, in which optimal or near-optimal controls may take the form of rapidly oscillating functions of time. This approach to OC of SP systems has also received considerable attention in the literature (see, e.g., Alvarez and Bardi [2], [3], [4], Artstein et al. [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], Borkar and

Gaitsgory [42], [43], [44], Donchev et al. [58], [59], [60], Filatov and Hapaev [72], Gaitsgory et al. [80], [81], [82], [83], [84], [85], [86], [87], [88], [89], [90], [91], [92], [93], [94], [95], [96], [97], Grammel [109], [110], [111], [112], [113], Leizarowitz [136], [137], Quincampoix et al. [152], [153], [154], Plotnikov et al. [150], [151], Vigodner [169], and references therein; further discussion of some of these contributions will be provided later in the text).

The LP approach to OC problems proves to be a suitable tool for dealing with SP systems. It is based on the fact that, under certain conditions, the optimal values of OC problems coincide with those of associated infinite-dimensional linear programming (IDLP) problems. In this book, we establish the validity of such LP representations for OC problems posed on the infinite time horizon—namely, for OC problems with time-discounting criteria and for OC problems with time-averaging criteria (periodic optimization problems), the latter playing a particularly important role in the analysis of SP control systems.

We show that the optimal values of the IDLP problems associated with these two classes of OC problems coincide with the optimal values of their dual problems (that is, strong duality holds), and we use these duality results to formulate sufficient and necessary optimality conditions for the OC problems under consideration. We further demonstrate that approximations of the IDLP problems by their semi-infinite-dimensional (SID) and finite-dimensional counterparts enable the numerical construction of near-optimal controls. Finally, we apply these LP techniques to the analysis of averaged OC problems that approximate the SP ones, and we show how asymptotically near-optimal controls for the latter can be constructed numerically.

Most of the results presented in this book have appeared in previously published papers. Bringing them together in a unified format is intended to provide the reader with a deeper and more comprehensive understanding of this area of research. Below is an overview of some key concepts and results discussed in the book.

(1) Singularly perturbed and reduced systems. Consider the controlled dynamical system defined by the equations

$$\epsilon \frac{dy(t)}{dt} = f(u(t), y(t), z(t)), \quad y(0) = y_0, \tag{1.1}$$

$$\frac{dz(t)}{dt} = g(u(t), y(t), z(t)), \quad z(0) = z_0, \tag{1.2}$$

where $f(\cdot)$ and $g(\cdot)$ are continuous vector functions that take values in $\mathbb{R}^m$ and $\mathbb{R}^n$ and satisfy Lipschitz conditions in (z, y) on a sufficiently large subset of $\mathbb{R}^m \times \mathbb{R}^n$ uniformly with respect to $u \in U$; U is a compact metric space, and the controls are the Lebesgue measurable functions taking values in U; and $\epsilon > 0$ is a small parameter (singular perturbation parameter).

The presence of the small parameter ϵ on the left-hand side of (1.1) implies that the rate of change of the y-components of the state variables is of order $O\left(\frac{1}{\epsilon}\right)$ and is therefore much higher than the rate of change of the z-components (due to the smallness of ϵ). Accordingly, the y- and z-components are referred to as fast and slow variables, respectively. Dynamical systems in which such a separation of time scales is induced by the singular perturbation parameter are called SP systems.

A natural idea when dealing with the SP system (1.1)–(1.2) is to set the singular perturbation parameter ϵ to zero, thereby converting (1.1) into a "static" equation:

$$f(u(t), y(t), z(t)) = 0. \tag{1.3}$$

Assuming that the equation

$$f(u, y, z) = 0 \tag{1.4}$$

has a unique root $y = \zeta(u, z)$, system (1.1)–(1.2) is then converted to the *reduced system* defined by the equation

$$\frac{dz(t)}{dt} = g(u(t), \zeta(u(t), z(t)), z(t)), \quad z(0) = z_0. \tag{1.5}$$

In the absence of controls, the approximation of the z-components of the SP system (1.1)–(1.2)) by the solutions of the reduced system (1.5) was established under natural stability conditions in the foundational works of Tikhonov [164] and Levinson [138]. However, this approximation may fail in the presence of controls that can take the form of rapidly oscillating functions of time.

(2) Averaged differential inclusion. Consider the so-called associated system

$$\frac{dy(\tau)}{d\tau} = f(u(\tau), y(\tau), z), \qquad z = \text{const}. \tag{1.6}$$

Note that the associated system looks similar to the "fast" subsystem (1.1) but (in contrast to the latter) it evolves in the "stretched" time scale $\tau = \frac{t}{\epsilon}$, with z being a vector of fixed parameters. The controls in (1.6) are the Lebesgue measurable functions taking values in U (as in the SP system (1.1)–(1.2)).

Throughout what follows, we assume that there exist sufficiently large compact sets $\hat{Y} \subset \mathbb{R}^m$ and $\hat{Z} \subset \mathbb{R}^n$ such that, for any $z \in \hat{Z}$, all solutions of the associated system that start in $\hat{Y}$ remain within the set $r\bar{B}^m$, where $\bar{B}^m$ denotes the closed unit ball in $\mathbb{R}^m$ and $r > 0$ is large enough. We also assume that, for any $z \in \hat{Z}$, there exists at least one initial condition $y(0) = y' \in \hat{Y}$ such that all solutions of the associated system that start at y' remain in $\hat{Y}$.

Denote by $V(z, S, y)$ the set of the time averages

$$V(z, S, y) \stackrel{\text{def}}{=} \bigcup_{u(\cdot)} \left\{ \frac{1}{S} \int_0^S g\big(u(\tau), y_z(\tau, u(\cdot), y), z\big) d\tau \right\}, \tag{1.7}$$

where the union is over all controls, with $y_z(\tau, u(\cdot), y)$ standing for the solution of associated system (1.6) obtained with a control $u(\cdot)$ and an initial condition $y_z(0) = y$.

Assume the following:

(i) *As the length of the time horizon S tends to infinity, the set $V(z, S, y)$ converges to a compact and convex set $V(z)$ for any $y \in \hat{Y}$ and $z \in \hat{Z}$. More precisely,*

$$\lim_{S\to\infty} d_H(V(z, S, y), V(z)) \le \nu_g(S) \quad \forall\, y \in \hat{Y}, \quad \forall\, z \in \hat{Z}, \tag{1.8}$$

where $\lim_{S\to\infty} \nu_g(S) = 0$ and $d_H(\cdot, \cdot)$ stands for the Hausdorff metric defined on bounded subsets of a finite-dimensional space by the Euclidean norm.

(ii) *The multivalued function $V(\cdot)$ is Lipschitz continuous.*

Under these assumptions, the z-components of the solutions of the SP system (1.1)–(1.2) are approximated by the solutions of the differential inclusion

$$\frac{dz(t)}{dt} \in V(z(t)), \quad z(0) = z_0, \tag{1.9}$$

which is referred to as the *averaged differential inclusion* (ADI).

Consider the differential inclusion

$$\frac{dz(t)}{dt} \in \bar{\text{co}} V_{ss}(z(t)), \quad z(0) = z_0, \tag{1.10}$$

where

$$V_{ss}(z) \stackrel{\text{def}}{=} \{v \mid v = g(u, \zeta(u,z), z), \quad u \in U\},$$

with $\zeta(u,z)$ being the root of the equation (1.4) and $\bar{co}$ standing for the closed convex hull of a set. Note that the set of solutions of the reduced system (1.5) is a subset of the set of solutions of the differential inclusion (1.10) and is dense in it.

From the fact that the set $V(z)$ is defined as the limit (1.8) and the fact that it is convex, it follows that

$$\bar{co}V_{ss}(z) \subset V(z), \tag{1.11}$$

with the inclusion (1.11) being proper in the general case (that is, $\bar{co}V_{ss}(z) \neq V(z)$). Therefore, the set of solutions of the differential inclusions (1.10) is a proper subset of the set of solutions of the ADI (1.9) in the general case. This explains why the z-components of some solutions of the SP system cannot be approximated by the solutions of the reduced system.

Under certain additional conditions, the limit set $V(z)$ is equal to the closure of its subset consisting of periodic regime points. A point $v \in V(z)$ is called a periodic regime point of $V(z)$ if there exist a positive number $\mathcal{T}$ and a control $u(\tau)$ defined on the interval $[0, \mathcal{T}]$ such that

$$v = \frac{1}{\mathcal{T}} \int_0^{\mathcal{T}} g(u(\tau), y_z(\tau), z) d\tau,$$

where $y_z(\tau)$ is the corresponding solution of the associated system that satisfies the periodicity condition $y_z(0) = y_z(\mathcal{T})$. This clarifies the fact that the use of rapidly oscillating controls, which induce oscillations in the fast state variables, is essential for obtaining adequate approximations of the slow state variables.

(3) Condition for the existence and convexity of the limit set $V(z)$. The assumption that the limit (1.8) exists, is convex, and does not depend on the initial conditions is crucial for establishing the fact that the z-components of the SP system (1.1)–(1.2) are approximated by the solutions of the ADI (1.9).

Let us introduce the following condition, which, for convenience, is referred to as the *mixing condition* in this introductory section:

For any $z \in \hat{Z}$, corresponding to any initial conditions $y', y'' \in \hat{Y}$ and any control $u'(\cdot)$, there exists a control $u''(\cdot)$ (which may or may not coincide with $u'(\cdot)$) such that the inequality

$$\left\| \frac{1}{S} \int_0^S g(u'(\tau), y_z(\tau, u'(\cdot), y'), z) d\tau - \frac{1}{S} \int_0^S g(u''(\tau), y_z(\tau, u''(\cdot), y''), z) d\tau \right\| \leq \nu_g(S) \tag{1.12}$$

is satisfied, where $\lim_{S\to\infty} \nu_g(S) = 0$.

It can be readily verified that, if the set $V(z,S,y)$ converges uniformly to a set $V(z)$ in the Hausdorff metric—that is, (1.8) holds—then the mixing condition is satisfied. In other words, this condition is necessary for the validity of (1.8). More interestingly, the mixing condition is also sufficient not only to guarantee the convergence of $V(z,S,y)$ to $V(z)$ but also to ensure the convexity of $V(z)$; that is, it implies the asymptotic convexification of the set $V(z,S,y)$ as S tends to infinity.

(4) Limit occupational measures set. The mixing condition is satisfied under certain stability and/or controllability assumptions discussed in the book. Moreover, these assumptions imply a stronger form of the mixing condition:

For any $z \in \hat{Z}$, corresponding to any initial conditions $y', y'' \in \hat{Y}$ and any control $u'(\cdot)$, there exists a control $u''(\cdot)$ (which may or may not coincide with $u'(\cdot)$) such that, for an arbitrary

continuous vector function $h(\cdot) \in \mathbb{R}^j$, $j = 1, 2, \ldots$, *the inequality*

$$\left\| \frac{1}{S} \int_0^S h(u'(\tau), y_z(\tau, u'(\cdot), y'), z) d\tau - \frac{1}{S} \int_0^S h(u''(\tau), y_z(\tau, u''(\cdot), y''), z) d\tau \right\| \leq \nu_h(S) \tag{1.13}$$

is satisfied, where $\lim_{S \to \infty} \nu_h(S) = 0$.

This stronger version of the mixing condition is sufficient to ensure that, for any continuous vector function $h(\cdot) \in \mathbb{R}^j$, $j = 1, 2, \ldots$, there exists a compact and convex set $V_h(z)$ such that the set $V_h(z, S, y)$,

$$V_h(z, S, y) \stackrel{\text{def}}{=} \bigcup_{u(\cdot)} \left\{ \frac{1}{S} \int_0^S h\big(u(\tau), y_z(\tau, u(\cdot), y), z\big) d\tau \right\},$$

converges in the Hausdorff metric to $V_h(z)$ as S tends to infinity:

$$\lim_{S \to \infty} d_H(V_h(z, S, y), V_h(z)) \leq \nu_h(S) \quad \forall\, y \in \hat{Y}, \quad \forall\, z \in \hat{Z}, \tag{1.14}$$

where $\lim_{S \to \infty} \nu_h(S) = 0$.

Given a control $u(\cdot)$ and the corresponding solution $y_z(\tau, u(\cdot), y)$ of the associated system (1.6) that satisfies the initial condition $y(0) = y$, denote by $\gamma_{z,S,y,u(\cdot)}$ the probability measure defined as

$$\gamma_{z,S,y,u(\cdot)}(Q) \stackrel{\text{def}}{=} \frac{1}{S} \int_0^S 1_Q(u(\tau), y_z(\tau, u(\cdot), y)) d\tau \quad \forall \quad \text{Borel } Q \subset U \times r\bar{B}^m, \tag{1.15}$$

where $1_Q(\cdot)$ stands for the indicator function of Q. This measure is referred to as the occupational measure generated by the control-state trajectory $(u(\cdot), y_z(\cdot))$ on the interval $[0, S]$. (Note that the aforementioned notation for the occupational measure differs slightly from the one used in the main body of the book.)

Denote by $\Gamma(z, S, y)$ the union of the occupational measures generated on the interval $[0, S]$ by the control-state trajectories of the associated system (1.6) that satisfy the initial condition $y(0) = y$. That is,

$$\Gamma(z, S, y) \stackrel{\text{def}}{=} \bigcup_{u(\cdot)} \{\gamma_{z,S,y,u(\cdot)}\}. \tag{1.16}$$

The convergence of $V_h(z, S, y)$ to a compact and convex set V_h for any continuous vector function $h(\cdot)$ implies that there exists a convex and weak* closed subset $\Gamma(z)$ of the space of probability measures on Borel subsets of $U \times r\bar{B}^m$ such that

$$\lim_{S \to \infty} \rho_H(\Gamma(z, S, y), \Gamma(z)) \leq \nu(S) \;\; \forall\, y \in \hat{Y} \quad \text{and} \quad \forall\, z \in \hat{Z}, \tag{1.17}$$

where $\lim_{S \to \infty} \nu(S) = 0$ and $\rho_H(\cdot, \cdot)$ is the Hausdorff metric on subsets of the space $\mathcal{P}(U \times r\bar{B}^m)$ defined by a metric compatible with the weak* topology on this space.[1]

The set $\Gamma(z)$, defined as the limit (1.17), is referred to as the *limit occupational measures set* (LOMS) of the associated system. Note that the LOMS $\Gamma(z)$ can be represented using the limits

[1] Here and everywhere in what follows, given a compact metric space $\mathcal{X}$, $\mathcal{P}(\mathcal{X})$ stands for the space of probability measures defined on Borel subsets of $\mathcal{X}$.

of the time averages $V_h(z)$, defined in (1.14), as follows:

$$\Gamma(z) \stackrel{\text{def}}{=} \Big\{ \gamma \in \mathcal{P}(U \times r\bar{B}^m) \,:\, \gamma(U \times \hat{Y}) = 1, \quad \int_{U\times\hat{Y}} h(u,y,z)\gamma(du,dy) \in V_h(z) \\ \forall \text{ continuous } h(\cdot) \in \mathbb{R}^j,\ j = 1,2,\dots \Big\}.$$

(The fact that $\gamma(U \times \hat{Y}) = 1$ for any $\gamma \in \Gamma(z)$ is implied by our assumption that, for at least one $y' \in \hat{Y}$, all solutions of the associated system starting at y' remain in $\hat{Y}$.)

Conversely, every limit of the time averages $V_h(z)$ can be represented in terms of the LOMS $\Gamma(z)$:

$$V_h(z) = \left\{ v \mid v = \int_{U\times\hat{Y}} h(u,y,z)\gamma(du,dy), \quad \gamma \in \Gamma(z) \right\}. \tag{1.18}$$

(5) Averaged control system. Setting $h(u,y,z) = g(u,y,z)$ in (1.18), we obtain the following representation for the set $V(z)$, which defines the right-hand side of the ADI (1.9):

$$V(z) = V_g(z) = \left\{ v \mid v = \int_{U\times\hat{Y}} g(u,y,z)\gamma(du,dy), \quad \gamma \in \Gamma(z) \right\}. \tag{1.19}$$

Thus, under mild conditions, the set of solutions of the ADI (1.9) coincides with the set of solutions of the control system

$$\frac{dz(t)}{dt} = \tilde{g}(\gamma(t), z(t)), \quad z(0) = z_0, \tag{1.20}$$

where the role of controls is played by the measure-valued functions $\gamma(\cdot)$ satisfying the inclusion

$$\gamma(t) \in \Gamma(z(t)), \tag{1.21}$$

with

$$\tilde{g}(\gamma, z) \stackrel{\text{def}}{=} \int_{U\times\hat{Y}} g(u,y,z)\gamma(du,dy). \tag{1.22}$$

The system (1.20)–(1.21) is referred to as the *averaged control system* (ACS). Note that, while—as mentioned above—the set of solutions to the ACS coincides with the set of solutions to the ADI, the former offers a different viewpoint on averaging and, in particular, enables the treatment of control systems with three time scales, as well as systems in which certain functions of the state variables (referred to as observables) evolve an order of magnitude more slowly than the state variables themselves.

(6) Stationary measures set. Assume for simplicity that there exists a compact set $Y \subseteq \hat{Y}$ which is forward invariant with respect to the solutions of the associated system for any $z \in \hat{Z}$; that is, any solution of the associated system that starts in Y remains in Y. Denote by $\Gamma(z,S,Y)$ the union of the occupational measures sets $\Gamma(z,S,y)$, defined in (1.16), over all initial conditions $y \in Y$:

$$\Gamma(z,S,Y) \stackrel{\text{def}}{=} \bigcup_{y\in Y} \{\Gamma(z,S,y)\}. \tag{1.23}$$

As the length of the time horizon S tends to infinity, the set $\bar{co}\Gamma(z,S,Y)$ (the closed convex hull of $\Gamma(z,S,Y)$) converges to a weak* compact and convex set $W(z)$ defined by

$$W(z) \stackrel{\text{def}}{=} \left\{ \gamma \in \mathcal{P}(U\times Y) : \int_{U\times Y} \nabla\phi(y)^T f(u,y,z)\gamma(du,dy) = 0 \ \forall\, \phi(\cdot) \in C^1(R^m) \right\} \tag{1.24}$$

and referred to as the *stationary measures set*. More specifically,

$$\rho_H(\bar{\text{co}}\Gamma(z,S,Y), W(z)) \le \nu_z(S), \tag{1.25}$$

where $\lim_{S\to\infty} \nu_z(S) = 0$.

If the LOMS $\Gamma(z)$ exists and is convex, then

$$\Gamma(z) = W(z).$$

Therefore, the ACS (1.20)–(1.21) takes the form

$$\frac{dz(t)}{dt} = \tilde{g}(\gamma(t), z(t)), \quad z(0) = z_0, \tag{1.26}$$

$$\gamma(t) \in W(z(t)). \tag{1.27}$$

(7) Important special case. Consider the SP system of the following special form:

$$\epsilon\frac{dy(t)}{dt} = f(u(t), y(t)), \quad y(0) = y_0, \tag{1.28}$$

$$\frac{dz(t)}{dt} = g(u(t), y(t)), \quad z(0) = 0, \tag{1.29}$$

where, unlike the general system (1.1)–(1.2), the right-hand sides are independent of z, and the slow dynamics is one-dimensional (that is, $g(\cdot) \in \mathbb{R}^1$). Note that, in this case, the associated system (1.6) and the stationary measures set (1.24) do not depend on z and are given respectively by

$$\frac{dy(\tau)}{d\tau} = f(u(\tau), y(\tau)), \tag{1.30}$$

and

$$W = \{\gamma \in \mathcal{P}(U\times Y) : \int_{U\times Y} \nabla\phi(y)^T f(u,y)\gamma(du,dy) = 0 \ \forall\, \phi(\cdot) \in C^1(\mathbb{R}^M)\}. \tag{1.31}$$

Accordingly, the ACS (1.26)–(1.27) takes the form

$$\frac{dz(t)}{dt} = \tilde{g}(\gamma(t)), \quad z(0) = z_0, \tag{1.32}$$

$$\gamma(t) \in W. \tag{1.33}$$

Consider the OC problem defined on the solutions of the SP system (1.28)–(1.29):

$$\inf_{u(\cdot)} z(1) = \inf_{u(\cdot)} \int_0^1 g(u(t), y(t))dt \stackrel{\text{def}}{=} G(\epsilon, y_0). \tag{1.34}$$

Consider also the OC problem defined on the solutions of the ACS (1.32)–(1.33):

$$\inf_{\gamma(\cdot)} z(1) = \inf_{\gamma(\cdot)} \int_0^1 \tilde{g}(\gamma(t))dt \stackrel{\text{def}}{=} \tilde{G}. \tag{1.35}$$

Provided that the LOMS of the associated system exists and is convex, the z-components of the solutions of the SP system (1.28)–(1.29) are approximated by the solutions of the ACS (1.32)–(1.33). This implies, in particular, that the optimal value of the former converges to the optimal value of the latter as ϵ tends to zero:

$$\lim_{\epsilon\to 0} G(\epsilon, y_0) = \tilde{G}. \tag{1.36}$$

The optimal control for problem (1.35) is constant valued:

$$\gamma(t) = \gamma^* \quad \forall\, t \in [0,1],$$

where γ^* is an optimal solution the "static" problem

$$\min_{\gamma \in W} z(1) = \min_{\gamma \in W} \tilde{g}(\gamma) \stackrel{\text{def}}{=} g^*, \tag{1.37}$$

with the optimal values of (1.35) and (1.37) being equal:

$$\tilde{G} = g^*.$$

Note that problem (1.37) belongs to the class of IDLP problems since both its objective function $\tilde{g}(\cdot)$ and the constraints defining the feasible set W are linear in the "decision variable" γ; see (1.22) and (1.31).

Note also that if there exists an optimal solution γ^* of the IDLP problem (1.37) that coincides with the occupational measure generated by a $\mathcal{T}$-periodic control-state trajectory $(u(\tau), y(\tau))$ of system (1.30), that is,

$$\gamma^*(Q) \stackrel{\text{def}}{=} \frac{1}{\mathcal{T}} \int_0^{\mathcal{T}} 1_Q(u(\tau), y(\tau)) d\tau \quad \forall \quad \text{Borel } Q \subset U \times Y,$$

then, provided that the system (1.30) satisfies certain stability conditions, the control $u(\frac{t}{\epsilon})$ will be asymptotically optimal in problem (1.34) in the sense that the value of the objective function obtained with this control will tend to g^* as ϵ tends to zero.

(8) IDLP problems related to periodic optimization and time-discounting OC problems. On the solutions of system (1.30), consider the OC problem

$$\inf_{(u(\cdot), y_0)} \frac{1}{S} \int_0^S g(u(\tau), y(\tau, u(\cdot), y_0)) d\tau \stackrel{\text{def}}{=} G_S, \tag{1.38}$$

where the minimization is over the controls and the initial conditions $y(0) = y_0 \in Y$. The optimal value of this problem is equal to the value of the problem of minimization over the occupational measures set $\Gamma(S, Y)$, defined analogously to (1.23):

$$G_S = \inf_{\gamma \in \Gamma(S,Y)} \int_{U \times Y} g(u,y) \gamma(du, dy) = \min_{\gamma \in \bar{co}\Gamma(S,Y)} \int_{U \times Y} g(u,y) \gamma(du, dy).$$

Therefore, by (1.25),

$$\lim_{S \to \infty} G_S = g^*, \tag{1.39}$$

where g^* is the optimal value of the IDLP (1.37).

Along with (1.38), consider the OC problem

$$\inf_{\mathcal{T}, (u(\cdot), y(\cdot))} \frac{1}{\mathcal{T}} \int_0^{\mathcal{T}} g(u(\tau), y(\tau)) d\tau \stackrel{\text{def}}{=} G_{per}\ , \tag{1.40}$$

where the minimization is over the length of the time interval $\mathcal{T}$ and over all control-state trajectories $(u(\cdot), y(\cdot))$ of system (1.30) that satisfy the periodicity condition $y(0) = y(\mathcal{T}) \in Y$. This problem is commonly referred to as the *periodic optimization problem*.

Assume that

$$\lim_{S\to\infty} G_S = G_{per}. \tag{1.41}$$

(This assumption is satisfied, for example, if the limit of the time averages set exists—that is, if (1.8) holds—and if the set of periodic regime points is dense in this limit.) Then, by (1.39), from (1.41) it follows that

$$G_{per} = g^*. \tag{1.42}$$

Note that, from (1.42), it follows that if $\mathcal{T}$ and $(u(\cdot), y(\tau))$ form an optimal solution of the periodic optimization problem (1.40), then the occupational measure generated by $(u(\cdot), y(\tau))$ on the interval $[0, \mathcal{T}]$ is an optimal solution of the IDLP problem (1.37).

Consider the family of IDLP problems depending on a parameter $C \geq 0$:

$$\min_{\gamma\in W(C,y_0)} \int_{U\times Y} g(u,y)\gamma(du,dy) \stackrel{\text{def}}{=} g^*(C,y_0), \tag{1.43}$$

where $y_0 \in Y$ and

$$W(C,y_0) \stackrel{\text{def}}{=} \Big\{\gamma \in \mathcal{P}(U\times Y) : \int_{U\times Y} (\nabla\phi(y)^T f(u,y) + C(\phi(y_0)-\phi(y)))\gamma(du,dy) = 0 \quad \forall \phi(\cdot) \in C^1(\mathbb{R}^M)\Big\}. \tag{1.44}$$

As can be seen,

$$W(0,y_0) = W \quad \text{and} \quad g^*(0,y_0) = g^*. \tag{1.45}$$

That is, the IDLP problem (1.37) is the member of this family corresponding to $C = 0$. As follows from the above discussion, this problem is related to the *long-run average* (LRA) OC and periodic optimization problems in the sense that (1.39) and (1.42) hold.

The members of the family (1.43) that correspond to $C > 0$ are related to the OC problem with time discounting, defined on the infinite time horizon

$$\inf_{u(\cdot)} \int_0^{+\infty} e^{-C\tau} g(u(\tau), y(\tau, u(\cdot), y_0))d\tau \stackrel{\text{def}}{=} V_C(y_0), \tag{1.46}$$

where C is the discount rate, and the minimization is over the controls and the corresponding solutions of system (1.30). Namely, for any $C > 0$, the optimal value of the IDLP problem (1.43) is equal to the C-multiple of the optimal value of the OC problem (1.46):

$$g^*(C,y_0) = CV_C(y_0) \quad \forall\, C > 0.$$

Further relationships between the aforementioned IDLP and OC problems are established through the duality result stated below. For any $C \geq 0$, the problem dual to the IDLP problem (1.43) can be written in the form

$$\sup_{\psi(\cdot)\in C^1(\mathbb{R}^m)} \min_{y\in Y} \{-H(y,\nabla\psi(y)) + C(\psi(y_0)-\psi(y))\} \stackrel{\text{def}}{=} \mu^*(C,y_0), \tag{1.47}$$

where the maximization is over all functions $\psi(\cdot) \in C^1(R^m)$, and $H(y,\xi)$ is the Hamiltonian:

$$H(y,\xi) \stackrel{\text{def}}{=} \max_{u\in U} \{-\xi^T f(u,y) - g(u,y)\}.$$

The duality result referred to above (commonly called *strong duality*) is the equality of the optimal values of (1.43) and its dual:

$$g^*(C, y_0) = \mu^*(C, y_0) \quad \forall\, C \geq 0. \tag{1.48}$$

Due to (1.48), one may conclude that a function $\psi(\cdot) \in C^1(\mathbb{R}^m)$ is an optimal solution of the dual problem if and only if

$$-H(y, \nabla\psi(y)) + C(\psi(y_0) - \psi(y)) \geq g^*(C, y_0) \quad \forall\, y \in Y. \tag{1.49}$$

This inequality represents a form of the celebrated Hamilton–Jacobi–Bellman (HJB) equation, and its solution, if it exists, can be used to state sufficient and necessary optimality conditions for the OC problems. More specifically, a $\mathcal{T}$-periodic control-state pair $(u(\tau), y(\tau)) = (u(\tau + \mathcal{T}), y(\tau + \mathcal{T}))$ is optimal for the periodic optimization problem (1.40) if and only if (1.49), taken with $C = 0$, holds as an equality along the trajectory $y(\cdot)$:

$$-H(y(\tau), \nabla\psi(y(\tau))) = g^*(0, y_0) = g^* = G_{per} \quad \forall\, \tau \in [0, \mathcal{T}]$$

(see (1.42) and (1.45)) and

$$y(\tau) = \operatorname{argmax}_{y \in Y}\{H(y, \nabla\psi(y))\} \quad \forall\, \tau \in [0, \mathcal{T}],$$

$$u(\tau) = \operatorname{argmax}_{u \in U}\{-\nabla\psi(y(\tau))^T f(u, y(\tau)) - g(u, y(\tau))\} \text{ a.e. } \tau \in [0, \mathcal{T}]. \tag{1.50}$$

Note that, due to (1.50), we may conclude that the control $u(y(\tau))$, where

$$u(y) = \operatorname{argmax}_{u \in U}\{-\nabla\psi(y)^T f(u, y) - g(u, y)\} \tag{1.51}$$

and where $y(\tau)$ is the periodic solution of the system

$$\frac{dy(\tau)}{d\tau} = f(u(y(\tau)), y(\tau)) \tag{1.52}$$

(assuming that such a periodic solution exists and is unique), will be optimal in the periodic optimization problem (1.40), provided that an optimal solution of the latter exists.

For the OC problem (1.46), a control $u(\tau)$ and the corresponding solution $y(\tau, u(\cdot), y_0) = y(\tau)$ of system (1.30) are optimal if and only if

$$-H(y(\tau, \nabla\psi(y(\tau))) + C(\psi(y_0) - \psi(y(\tau))) = g^*(C, y_0) = CV_C(y_0) \quad \forall\, \tau \in [0, \infty),$$

$$y(\tau) = \operatorname{argmax}_{y \in Y}\{H(\nabla\psi(y), y) + C\psi(y)\} \quad \forall\, \tau \in [0, \infty),$$

$$u(\tau) = \operatorname{argmax}_{u \in U}\{-\nabla\psi(y(\tau))^T f(u, y(\tau)) - g(u, y(\tau))\} \text{ a.e. } \tau \in [0, \infty). \tag{1.53}$$

Also, by (1.53), the control $u(y(\tau))$, where $u(y)$ is defined in (1.51) and $y(\tau)$ is the solution of (1.52) with the initial condition $y(0) = y_0$ (assuming it exists and is unique), will be optimal in the OC problem (1.46), provided that an optimal solution of this problem exists.

(9) Semi-infinite-dimensional approximations of the IDLP problems. Let $\{\phi_i(\cdot) \in C^1(\mathbb{R}^m)\, , i = 0, 1, \dots\}$ be a sequence of functions such that any function $\psi(\cdot) \in C^1(\mathbb{R}^m)$ and its gradient $\nabla\psi(\cdot)$ can be simultaneously approximated on Y by linear combinations of functions from $\{\phi_i(\cdot)\, , i = 1, 2, \dots\}$ and their corresponding gradients. Consider the family of *semi-infinite-dimensional* linear programming (SIDLP) problems

$$\inf_{\gamma \in W_N(C, y_0)} \int_{U \times Y} g(u, y)\gamma(du, dy) \stackrel{\text{def}}{=} g_N^*(C, y_0), \tag{1.54}$$